

Predicting Mouse Ultrasonic Vocalization Type from Pre-Call Neural Population Activity

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Abstract

Mouse ultrasonic vocalizations (USVs) vary in duration and frequency modulation during social interactions. We ask whether neural population activity before a vocalization contains information about the acoustic type of the upcoming call. Because no ground-truth call-type labels are available, calls are first filtered by acoustic clarity and represented using measurable acoustic features. K-Means then identifies three stable, data-driven acoustic categories. Neural features are constructed from population spike-count activity during the second preceding each call and evaluated with temporally blocked, bout-grouped cross-validation and a circular-shift null model. Only one of 21 session-by- K decoding tests survives FDR correction, indicating limited and session-specific evidence for pre-call call-type information. A positive-control analysis detects a broader vocalization-related neural signal when evidence is pooled across sessions, helping distinguish weak type-specific decoding from a pipeline that is insensitive to vocalization-related activity altogether.

1. Introduction

Mice produce ultrasonic vocalizations (USVs) during social interactions, including courtship-related contexts [1]. In the larger experiment used here, the recorded subject mice were male and interacted with both male and female mice. Vocal activity was very uneven across sessions: almost all usable calls in the analyzed data came from the female-encounter recordings of M7 and M8. We therefore focus the acoustic analysis on the 16 female-encounter sessions and later restrict neural decoding to the sessions that contain enough usable calls.

Our main question is: *given that a mouse produces a vocalization, can neural activity before the call predict the acoustic type of that call?* Mouse USV structure is related to behavior: Sangiamo et al. reported associations between vocal patterns, social actions, and pitch trajectories [1]. This motivates asking not only whether a call occurs, but whether the neural state before the call is related to the acoustic form that follows.

The data contain no manual ground-truth syllable labels, so call types must first be defined from the audio. This is similar in motivation to MUPET, which uses unsupervised analysis of spectro-temporal USV structure [2]. Such labels still require caution: Hertz et al. showed that USV labels can depend on the labeling approach and that calls have temporal sequence structure [3]. We therefore treat our clusters as operational acoustic categories rather than fixed biological syllable classes.

Previous work also connects neural activity to mouse vocal behavior. Tschida et al. showed that neurons in the *periaqueductal gray* (PAG), a midbrain region involved in vocal production, can affect whether social USVs are produced [5]. Gan-Or and London reported increased activity in the anterior cingulate cortex (ACC) before vocal initiation [6], while Chen et al. showed that neural circuits can influence properties of vocal output [4]. In this project we therefore do two things: first, define reliable acoustic call categories from the audio; second, go back to the neural activity before each call and test whether it helps predict which acoustic category is about to occur.

2. Problem Formulation

2.1 Acoustic Call Representation

For each usable call i , define

$$(1) \quad \mathbf{a}_i = [d_i, f_i, s_i]^\top,$$

where d_i is duration, f_i is the intercept of a linear fit to the tracked frequency contour (a starting-frequency summary), and s_i is its slope. Each feature j is standardized,

$$(2) \quad \tilde{a}_{ij} = \frac{a_{ij} - \mu_j}{\sigma_j},$$

because K-Means uses Euclidean distance and the variables have different numerical scales. For K clusters, the label is

$$(3) \quad z_i = \arg \min_{k \in \{1, \dots, K\}} \|\tilde{\mathbf{a}}_i - \boldsymbol{\mu}_k\|_2^2,$$

where $\boldsymbol{\mu}_k$ is centroid k .

2.2 Neural Prediction Problem

Once each usable call has an acoustic label, its onset time becomes the reference point for the neural analysis. For every call, we move backward by T seconds and examine the neural population activity leading up to that onset. Rather than averaging the whole pre-call period into one value per neuron, we divide it into B shorter consecutive bins. Each neuron therefore contributes one activity value per bin, which keeps a coarse description of *when* activity changes as the call approaches.

Let t_i be the onset of call i . The pre-call window is

$$(4) \quad \mathcal{W}_i = [t_i - T, t_i),$$

and if $x_{i,c,b}$ is the mean spike-count activity of neuron c in bin b , the neural feature vector is

$$(5) \quad \mathbf{x}_i = [x_{i,1,1}, \dots, x_{i,C,B}]^\top.$$

The primary setting uses $T = 1$ s and $B = 2$, so the second before the call is represented by two 0.5-s bins and a session with C cells gives $2C$ neural features. We then learn

$$(6) \quad \hat{z}_i = f_\theta(\mathbf{x}_i),$$

where f_θ is the training-fold pipeline of standardization, PCA, and classification. The test is whether these pre-call features predict the acoustic label better than expected when the real temporal alignment between neural activity and labels is broken.

3. Methods

3.1 Data and Temporal Organization

The full experiment recorded mice during interactions with both male and female mice. The present analysis uses 16 female-encounter sessions from five recorded mice, with 3,984 detected vocalizations. Nearly all vocalizations that remain usable after acoustic quality control except for a small number of sessions: 2,656 of the 2,692 usable calls (98.7%) come from M7 and M8, and the seven sessions that later enter neural decoding are all from these two mice. The complete session inventory and decoding eligibility are reported in Appendix A.1. The male-encounter recordings remain part of the larger experiment but contribute too few usable

calls to the present call-type decoding analysis. Neural activity is supplied as `spikeCounts100ms`, while acoustic features are measured from the raw .wav recordings. Following the course-staff instructions, processed vocalization timestamps are mapped to the raw audio using the session-specific `vocSyncTime` correction. The synchronization correction and its audio-energy sanity check are detailed in Appendix A.2.

Calls are grouped into bouts because nearby calls can have related labels and overlapping pre-call neural history. The inter-call interval distribution has no clear separating valley, so the bout boundary is obtained from the knee of the number-of-bouts curve, giving a 0.4-s silence threshold. Calls belonging to the same bout are kept together during validation. Audio/background and contour-processing details are provided in Appendix A.3, while the bout construction and threshold are shown in Appendix A.8.

3.2 Acoustic Processing, Clarity, and Call Types

Each detected timestamp is treated as an approximate call location rather than as a complete acoustic measurement. We compute a 30–110 kHz spectrogram from the raw audio, estimate a separate quiet-period background for each recording, and track the strongest continuous frequency ridge around the detection. Duration and frequency features are then computed only from the accepted contour. Session-specific normalization is important because the recordings contain different stationary spectral peaks and noise floors; examples and the full procedure are shown in Appendix A.3. Bandpass and stationary-tone sanity checks are reported in Appendix A.4.

3.2.1 Call clarity

A detected vocalization is not automatically suitable for defining a call type. If the contour is weak, cut by the extraction window, or cannot be tracked reliably, its measured duration and frequency trajectory may mainly reflect tracking error. We therefore filter in two steps. First, a call must pass structural contour checks: a reliable anchor must be found, the contour must not be edge-truncated, and it must contain enough valid points. Second, for calls that can be tracked, we measure how consistently the contour rises above the quiet-period noise level. Define:

$$(7) \quad q_i = \frac{\#\{\text{tracked frames with peak energy} > 1.5P_{99.9}^{\text{quiet}}\}}{\#\{\text{tracked frames}\}},$$

where $P_{99.9}^{\text{quiet}}$ is the 99.9th percentile of peak in-band energy measured during a quiet period in the same session. We use this extreme upper-tail value as a conservative quiet-noise reference so that ordinary background fluctuations are unlikely to count as strong call frames. It was not selected by optimizing the final decoding result. A diagnostic showing where this reference lies relative to quiet and call-frame energy is provided in Appendix A.5.

Table 1: Call-clarity levels and primary inclusion rule.

Category	Criterion	N	Used?
Clear	$q_i \geq 0.60$	1,291	Yes
Partially Clear	$0.30 \leq q_i < 0.60$	1,401	Yes
Unclear/unusable	$q_i < 0.30$ or QC fail	1,292	No

The primary clustering sample is therefore exactly Clear $\cup$ Partially Clear: 2,692 calls, or 67.6% of all detections. There is no additional selection after this clarity step. Detailed exclusion reasons and session-wise clarity composition are reported in Appendix A.6. The stricter Clear-only subset and the set of every otherwise valid tracked call are used later only to test sensitivity to this cutoff (Appendix B.10).

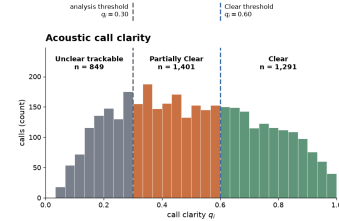


Figure 1: Call clarity. Histogram of q_i for the 3,541 calls with a trackable contour. The 0.30 line is the primary inclusion threshold and 0.60 separates Clear from Partially Clear calls. The 443 detections that fail structural contour QC have no valid q_i and are included only in the Unclear/unusable total in Table 1.

3.2.2 Feature selection

Five acoustic variables were available for clustering: duration, FM intercept, FM slope, minimum frequency, and maximum frequency. We chose the feature set before selecting K so that the cluster definition would not be tuned after seeing which partition looked best. The correlation analysis showed that the three frequency-level measures contain substantial overlap: FM intercept correlates with minimum frequency at $r = 0.87$ and with maximum frequency at $r = 0.88$, while minimum and maximum frequency correlate at $r = 0.86$. In K-Means, every standardized feature contributes its own squared term to Euclidean distance. Keeping all three correlated frequency measures would therefore count similar frequency information several times and could make frequency dominate the partition.

We consequently keep {duration, FM intercept, FM slope}. These variables describe complementary aspects of a call: how long it lasts, its overall starting-frequency level, and the direction/rate at which frequency changes. The full correlation matrix is shown in Appendix B.1. We also report leave-one-feature-out clustering and pairwise cluster comparisons in Appendix B.9 and Appendix B.6, respectively, to show how strongly each selected variable contributes to the final partition. Sensitivity to retaining the original five-feature representation is reported in Appendix B.2.

3.2.3 Choice of K and clustering model

Because there are no ground-truth syllable labels, K is selected using internal cluster-validation measures. The **Silhouette score** compares within-cluster cohesion with separation from the nearest competing cluster; larger values indicate a clearer partition [7]. We use it as the primary criterion. As supporting checks, the **Davies–Bouldin index** compares within-cluster scatter with between-cluster separation, where lower is better [8], and the **Calinski–Harabasz index** is a between- versus within-cluster variance ratio, where higher is better [9]. Across $K = 2, \dots, 7$, Silhouette and Calinski–Harabasz favor $K = 3$, while Davies–Bouldin favors the finer $K = 7$ partition. We therefore select $K = 3$ according to the predefined primary criterion; the full sweep is shown in Appendix B.3.

The three internal indices are used to choose K for the K-Means partition, not to decide whether K-Means is intrinsically better than a GMM. In particular, Silhouette is distance-based and tends to reward compact, well-separated Euclidean clusters, a geometry that is naturally close to the squared-Euclidean objective optimized by K-Means. We therefore do not use Silhouette as a head-to-head K-Means-versus-GMM score. Instead, GMM model form is evaluated with BIC and the resulting partitions are compared with ARI.

We fit both diagonal- and full-covariance Gaussian mixture models (GMMs). A diagonal GMM estimates one variance per feature and cluster but assumes zero within-cluster covariance; a full GMM also models pairwise covariance. At $K = 3$, the full-covariance model has lower BIC than the diagonal model (17,449 vs. 18,246), so its additional flexibility improves the likelihood enough to justify the extra GMM parameters. At $K = 3$ and $d = 3$ acoustic features, K-Means estimates $Kd = 3 \times 3 = 9$ centroid coordinates. A

full-covariance GMM estimates $Kd = 9$ mean parameters, $Kd(d + 1)/2 = 18$ independent covariance parameters, and $K - 1 = 2$ free mixture weights, giving $9 + 18 + 2 = 29$ free parameters in total; the diagonal GMM uses 9 means, 9 variances, and 2 free weights, giving 20 parameters. The full parameter accounting and model-complexity comparison are given in Appendix B.4. The K-Means and full-GMM partitions show substantial but not identical agreement ($\text{ARI} = 0.693$). The **Adjusted Rand Index (ARI)** measures agreement between partitions while correcting for chance and ignoring arbitrary cluster numbering [10]. Since the downstream decoder needs one hard target label per call, and the more complex GMM does not produce a qualitatively different partition, we carry K-Means forward as the more parsimonious model. Alternative PCA, t-SNE, and UMAP visualizations of the K-Means and GMM labelings are shown in Appendix B.8.

Repeated 80% subsampling gives mean $\text{ARI} = 0.98$ at $K = 3$, indicating that the selected partition is stable when the sampled calls change. Full subsampling-stability and acoustic null-model diagnostics are reported in Appendix B.5.

3.3 Neural Prediction

Only sessions with at least 100 usable labeled calls enter decoding: M7.F1–F4 and M8.F2–F4. Session-wise call counts and $K = 3$ class balance are given in Appendix C.1. For each call, the 1 s before onset is split into two 0.5-s bins, and every recorded neuron contributes its mean `spikeCounts100ms` value in each bin. The split retains a simple temporal contrast: two calls can have similar total pre-call activity while differing in whether neural activity changes as call onset approaches.

The neural feature dimension is large relative to the number of events, so standardization and PCA are fitted inside each training fold only. We compare regularized logistic regression, random forest, and RBF-SVM using nested cross-validation; hyperparameter tuning is performed only inside the training data so the held-out folds remain independent of model selection. Mean Macro-F1 is 0.359 for logistic regression, 0.328 for random forest, and 0.320 for RBF-SVM; logistic regression wins 4 of 7 sessions and is carried forward. The most frequently selected configuration is $C = 0.01$ with 20 PCA components (full comparison in Appendix C.2). After model comparison, this logistic-regression pipeline is fixed as the primary decoder; the same preprocessing, PCA configuration, classifier, and hyperparameters are then applied to the observed labels and to every circular-shift null draw.

3.4 Validation and Statistical Testing

Random K-fold splitting can place calls from the same bout in both training and test sets, even though these calls may share labels and overlapping neural history. We therefore use five temporally blocked folds grouped by bout, with a temporal exclusion margin around each held-out block. This choice is not only conservative in principle: the intentionally invalid random-CV comparison increases Macro-F1 by 0.056 on average, with inflation up to 0.134 (Appendix D.4).

For class k , let TP_k , FP_k , and FN_k denote true positives, false positives, and false negatives. Precision Pre_k and recall Rec_k are

$$(8) \quad Pre_k = \frac{TP_k}{TP_k + FP_k}, \quad Rec_k = \frac{TP_k}{TP_k + FN_k},$$

and the primary score is

$$(9) \quad F1_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \frac{2 Pre_k Rec_k}{Pre_k + Rec_k}.$$

Macro-F1 gives each acoustic class the same weight, which is important because several sessions have imbalanced cluster counts.

For the primary $K = 3$ task, session-wise Macro-F1 ranges from 0.272 to 0.406 (mean 0.346); M7.F2 reaches 0.392 and is the only session that later survives the full significance correction. Accuracy and majority-class accuracy are reported descriptively, while significance is determined from the empirical circular-shift null rather than from a theoretical chance value.

Statistical significance is evaluated with a circular-shift null. For example, if the time-ordered labels are $[1, 1, 2, 3, 3]$, a shift by two positions gives $[3, 3, 1, 1, 2]$ while the neural samples stay fixed. This breaks the true pairing between a call’s neural activity and its label but keeps the label sequence and its temporal autocorrelation intact, unlike a fully random permutation. With $B = 1000$ shifts,

$$(10) \quad p = \frac{1 + \#\{b : S_b^{\text{null}} \geq S_{\text{obs}}\}}{B + 1}, \quad B = 1000,$$

where S_{obs} is the observed Macro-F1 and $\#\{b : S_b^{\text{null}} \geq S_{\text{obs}}\}$ is simply the number of shifted-label runs whose score is at least as large as the observed score. Families of session/configuration tests are corrected using Benjamini–Hochberg FDR, with $q < 0.05$ considered significant.

3.5 Control Analyses and Their Purpose

A weak or negative decoding result has two possible explanations: the target information may truly be weak, or the analysis may fail to detect a signal that is actually present. Positive controls are used to distinguish these possibilities by checking whether the same experimental or analysis system can recover an effect that is expected to be easier to detect. G6rgen et al. formalized this idea for multivariate analysis as the “Same Analysis Approach,” recommending that the same pipeline be applied to control data rather than changing methods for the control [11]. More generally, positive controls are recommended when interpreting negative findings because they help reveal false negatives caused by an insensitive setup or analysis [12].

Our main positive control is therefore **calling versus quiet**. The encounter is divided into non-overlapping 0.5-s windows and the same neural pipeline predicts whether each window contains a vocalization. This is deliberately a simpler target than predicting acoustic call type: it asks whether the neural data contain detectable information about vocalization state at all. If this control were also negative, a negative call-type result would be harder to distinguish from insufficient sensitivity of the analysis pipeline. We additionally test **post-call call type**, using the same acoustic labels but neural activity immediately after call onset. This second analysis is complementary rather than a required positive control: it asks whether category-related neural information is easier to detect around production than before it. Both analyses use the same preprocessing, classifier, blocked cross-validation, circular null, and FDR framework as the primary analysis. Complete per-session control results are reported in Appendix C.5, and the secondary pooled analysis across recordings is reported in Appendix C.6.

4. Results

4.1 Acoustic Quality and Call Types

Of 3,984 detections, 1,291 (32.4%) are Clear, 1,401 (35.2%) Partially Clear, and 1,292 (32.4%) Unclear/unusable. The 2,692 calls used to define the acoustic categories are therefore exactly the first two clarity groups. Re-clustering the stricter Clear-only subset still agrees well with the primary labels ($\text{ARI} = 0.861$), while re-clustering every otherwise valid tracked call gives $\text{ARI} = 0.987$. Thus the three-category structure is not created by selecting a narrow quality subset, although the retained-versus-excluded comparison in Appendix A.7 shows that acoustic filtering is not completely neutral with respect to every acoustic feature.

Table 2: Median characteristics of the $K = 3$ acoustic categories.

Cl.	N	Dur. (ms)	Int. (kHz)	Slope (kHz/s)
0	757	45.5	62.52	+16.67
1	994	18.0	60.70	+459.59
2	941	43.0	43.26	-75.94

The selected $K = 3$ solution has Silhouette = 0.394, 80% subsampling ARI = 0.98, and cluster sizes 757 (28%), 994 (37%), and 941 (35%). Cluster 0 is long, relatively flat, and high-frequency; Cluster 1 is short, strongly rising, and high-frequency; Cluster 2 is long, falling, and lower-frequency. The pooled clustering sample is dominated by M7 (84.4%), with M8 contributing 14.3% and the remaining mice about 1.3% combined, which limits how strongly the categories can be generalized across animals. The full stability and null-model validation is reported in Appendix B.5, detailed acoustic cluster profiles are given in Appendix B.6, and mouse-wise cluster composition is shown in Appendix B.7.

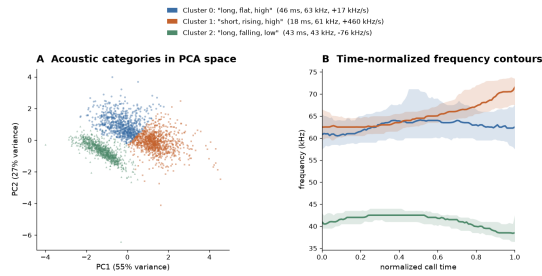


Figure 2: Data-driven acoustic categories. Panel A shows the selected $K = 3$ K-Means labels in a descriptive two-dimensional PCA projection of the standardized clustering features. Panel B shows the median time-normalized frequency contour for each category, with the interquartile range shaded, using up to 120 calls per cluster. Clustering itself is performed in the original standardized three-feature space, not in the PCA projection. Full K -selection diagnostics are shown in Appendix B.3; the K-Means/GMM complexity comparison and alternative embeddings are reported in Appendices B.4 and B.8.

4.2 Prediction of Upcoming Call Type

The main neural analysis evaluates independently fitted targets at $K = 2, 3, 4$ across seven eligible sessions. Only 1 of the 21 session-by- K tests survives BH-FDR correction:

$$(11) \quad K = 3, M7.F2: \quad F1_{\text{macro}} = 0.392, \quad p = 0.002, \quad q = 0.042.$$

The other $K = 3$ Macro-F1 values are 0.333, 0.370, 0.406, 0.359, 0.291, and 0.272 across the remaining sessions. The complete 21-test $K = 2, 3, 4$ sweep is reported in Appendix C.3. M7.F2 therefore provides a session-specific positive result, but the effect is not reproduced consistently. The data do **not** support a dataset-wide conclusion that upcoming acoustic type is reliably decodable from pre-call population activity.

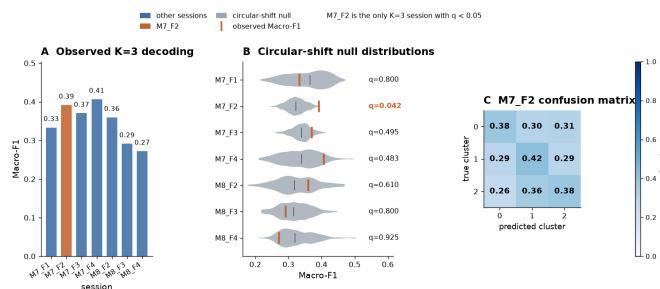


Figure 3: Pre-call decoding of acoustic type. Panel A shows observed $K = 3$ Macro-F1 across the seven decoding-eligible sessions. Panel B shows the 1,000-draw circular-shift null distribution for each session with the observed score marked and its BH-FDR q value reported. Panel C shows the row-normalized confusion matrix for M7.F2. M7.F2 is the only $K = 3$ session that survives correction.

4.3 Control Analyses: Can the Pipeline Detect Vocalization-Related Activity

Neither control survives FDR correction at the individual-session level (0/7 for calling-versus-quiet and 0/7 for post-call type). The complete per-session control results are given in Appendix C.5. Because each recording contains only a few hundred events, we also combine session-level p -values as a clearly labeled secondary analysis across recordings. Calling-versus-quiet is then significant (Fisher $p = 0.0024$, Stouffer $p = 0.0008$, FDR $q = 0.0047$), whereas post-call call-type decoding remains non-significant ($q = 0.2037$). The pooled pre-call $K = 3$ result also remains non-significant after correction ($q = 0.080$). Because several recordings come from the same animals, this pooled result is a recording-level summary rather than independent animal-level replication. The complete pooled-evidence table is reported in Appendix C.6.

This control changes how the weak main result should be interpreted. The same neural-analysis pipeline can detect a broad vocalization-related signal when information is combined across recordings, so the lack of robust call-type decoding is not simply a case where the pipeline detects no vocalization-related information at all. At the same time, the positive control does not prove that each individual session has enough statistical power to detect a weaker type-specific signal.

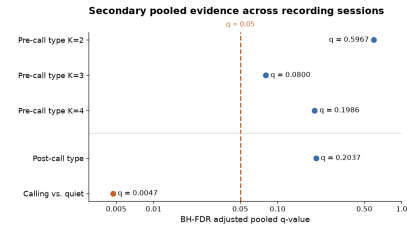


Figure 4: Secondary pooled evidence across recordings. Session-level circular-shift p -values are combined across recording sessions using Fisher's method. BH-FDR correction is then applied separately across the three pre-call call-type targets and across the two control analyses. Points show the resulting adjusted q -values on a logarithmic scale; the dashed line marks $q = 0.05$. Only the calling-versus-quiet control survives correction ($q = 0.0047$). Because several recordings come from the same animals, these values summarize evidence across recordings and are not independent animal-level replications.

4.4 Robustness

The conclusion remains stable across pre-call windows of 0.5, 1, and 2 s: only one of 21 window-by-session tests is significant after correction, and the 1-s window has the highest mean Macro-F1 (full results in Appendix C.4). Changing the bout threshold from 0.4 s to 0.3 or 0.6 s also leaves the qualitative result unchanged (Appendix D.3). Region and minimum-class-size checks are reported in Appendix D.1 and Appendix D.2, respectively. Acoustic feature ablation and clarity-threshold sensitivity are reported in Appendix B.9 and Appendix B.10, while the grouped-versus-random-CV leakage check is reported in Appendix D.4.

5. Discussion

The acoustic analysis provides a stable three-category prediction target. Across the three variables used to construct the clusters, all pairwise cluster comparisons are statistically significant after Holm correction, although the effect sizes range from negligible to large (Appendix B.6). Because these same variables were used to form the clusters, the pairwise tests are descriptive rather than independent validation. More importantly, the partition remains stable under repeated 80% subsampling (Appendix B.5) and under stricter or looser call-clarity thresholds (Appendix B.10).

The neural result is more limited. M7.F2 contains significant pre-call information for the $K = 3$ target, but this effect is not reproduced consistently across the other recordings or across the broader K sweep. The full cross- K results are given in Appendix C.3. The

strongest conclusion is therefore that the dataset contains *limited, session-dependent* evidence that pre-call population activity predicts the acoustic type of the upcoming call. Descriptive pre-call PSTHs in Appendix C.7 help separate this result from simple changes in the population mean: the significant M7_F2 session actually has more compact cluster-average traces than the non-significant M7_F1 session. This is neither better nor worse for multivariate decoding, because the PSTH averages across neurons whereas the classifier can use distributed cell-specific patterns that partly cancel in a population average. The calling-versus-quiet control strengthens the interpretation of the weak main result: when evidence is pooled across sessions (Appendix C.6), the same pipeline can detect a broad vocalization-related neural signal, so the main result is not simply a case where the analysis detects no vocalization-related information at all.

5.1 Limitations

The acoustic labels are data-driven rather than ground-truth syllable labels, and their interpretation depends on the chosen representation and labeling procedure; this general issue has also been emphasized for mouse USV labeling by Hertz et al. [3]. The acoustic sample is strongly dominated by M7 (Appendix B.7), the recorded brain-region composition differs between M7 and M8 (Appendix D.1), and caller identity is not known for each vocalization. In addition, `spikeCounts100ms` is already temporally smoothed before our 0.5-s binning, which may hide faster pre-call changes.

5.2 Future Work

The most direct next step is to collect more recordings with larger numbers of usable vocalizations, especially sessions with enough examples from every acoustic category. More calls would increase the statistical power of the circular-null tests and reduce the sensitivity of Macro-F1 to very small classes. Improving audio quality and caller identification would also reduce uncertainty in the acoustic labels, while neural features with finer temporal resolution could test whether useful pre-call changes occur on a shorter timescale than the current representation captures.

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Supplementary Material

A. Data and Acoustic Quality Control

A.1 Session Inventory

The larger experiment contains interactions with both male and female mice. The present analysis uses the 16 female-encounter sessions because nearly all acoustically usable vocalizations are concentrated in a few of these recordings, especially M7 and M8. Table S1 gives the complete female-session inventory used here.

Table S1: Analyzed female-encounter session inventory. A session enters neural decoding only if at least 100 acoustically usable labeled calls remain after quality control.

Session	Detected calls	Usable calls	Recorded cells	Stimulus	Neural decoding
M2_F1	14	2	114	Female	No
M2_F2	8	1	136	Female	No
M2_F3	0	0	118	Female	No
M3_F1	0	0	68	Female	No
M3_F2	2	1	63	Female	No
M3_F3	0	0	62	Female	No
M3_F4	5	3	68	Female	No
M6_F1	32	29	233	Female	No
M7_F1	779	574	166	Female	Yes
M7_F2	941	727	177	Female	Yes
M7_F3	1125	841	163	Female	Yes
M7_F4	134	129	212	Female	Yes
M8_F1	1	1	108	Female	No
M8_F2	370	119	130	Female	Yes
M8_F3	362	136	98	Female	Yes
M8_F4	211	129	139	Female	Yes
Total	3984	2692	–	–	7 sessions

A.2 Audio-Time Correction

Processed vocalization times and raw audio are stored on different time references. Following the course staff instructions, the session-specific vocSyncTime correction is applied whenever raw audio is extracted around a processed call time. As a sanity check, USV-band energy around corrected call times was compared with the same analysis using no correction and the opposite sign. The instructed correction gave the largest call-to-control energy ratio in all 8 sessions with enough calls for this diagnostic (median corrected ratio 5.0 $\times$, compared with 1.1 $\times$ with no offset and 1.0 $\times$ with the sign flipped).

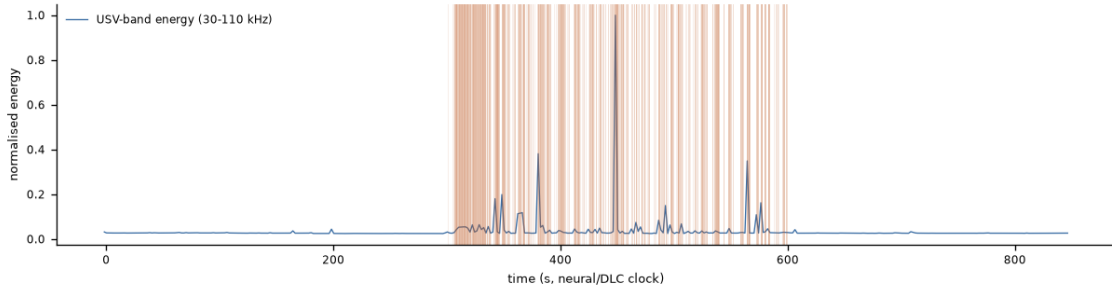


Figure S1: Representative audio-time alignment after applying the instructed session-specific correction. The three-way corrected/no-offset/sign-flipped comparison is reported numerically in Table S2.

Table S2: Audio-time correction sanity check. Values are call-to-control USV-band energy ratios; the instructed correction is largest in all eight sessions with enough calls for the diagnostic.

Session	M6_F1	M7_F1	M7_F2	M7_F3	M7_F4	M8_F2	M8_F3	M8_F4
Corrected	80.42	7.18	4.93	4.99	83.66	3.60	3.05	4.11
No offset	1.04	1.08	1.05	1.18	1.07	1.05	1.09	1.01
Sign flipped	1.04	1.07	1.02	1.15	1.05	1.05	1.04	0.99

A.3 Background Noise and Contour Extraction

Background spectra differ across recordings: the strongest stationary spectral peak ranges from 47.9 to 68.6 kHz across 15 sessions with detected peaks. Because no single notch frequency is appropriate for all sessions, a quiet-period background spectrum is estimated and subtracted separately for each recording. The ridge tracker then operates on this session-normalized spectrogram.

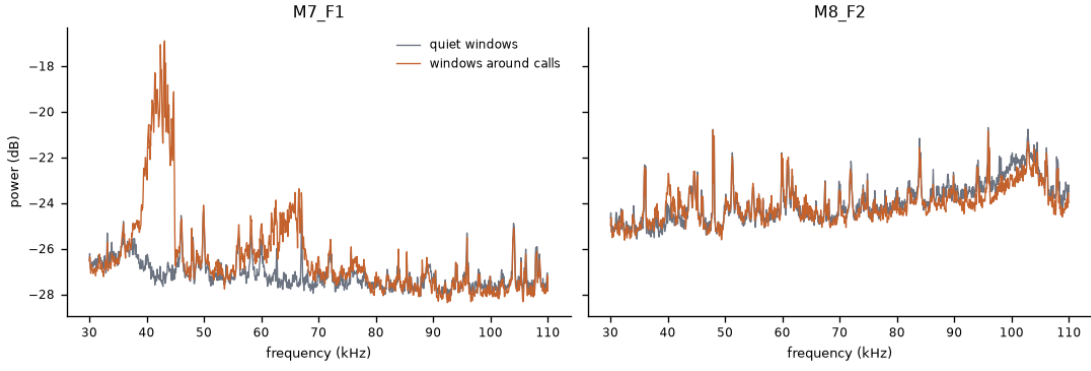


Figure S2: Session-specific background structure. Stationary spectral energy is estimated from quiet periods and removed before contour tracking.

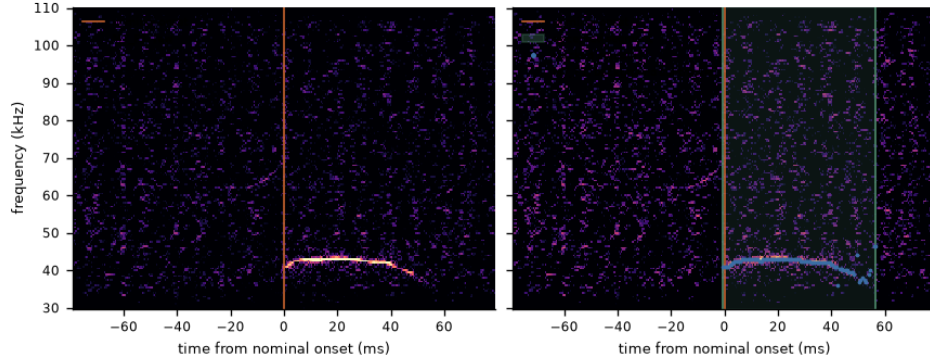


Figure S3: Example of background normalization and contour extraction. Acoustic features are computed only from the accepted tracked contour.

A.4 Bandpass and Static-Tone Sanity Checks

The 30–110 kHz band does not visibly cut into the retained calls: only 11 of 2,692 usable calls reach within 3 kHz of the 30-kHz lower edge and none reach within 3 kHz of the 110-kHz upper edge. We also checked whether session-specific stationary tones were being mistaken for calls. Across the seven decoding sessions, the ratio of calls peaking within 1 kHz of the strongest stationary tone relative to the surrounding 10-kHz neighborhood ranges from 0.00 to 0.97; values substantially above 1 would be expected if the static tone itself were being tracked as a call.

Table S3: Static-tone sanity check in the seven decoding sessions.

Session	Tone (kHz)	Calls within 1 kHz (%)	Ratio vs. local
M7_F1	67.0	5.7	0.74
M7_F2	50.5	0.3	0.77
M7_F3	66.5	8.0	0.97
M7_F4	66.0	0.8	0.42
M8_F2	47.9	0.0	0.00
M8_F3	48.1	0.0	0.00
M8_F4	47.9	3.1	0.77

A.5 Quiet-Period Reference Percentile

The clarity score uses $P_{99.9}^{\text{quiet}}$, the 99.9th percentile of the maximum in-band energy observed in a quiet reference period after the same background normalization used for calls. This percentile was used as a conservative upper-tail noise reference rather than selected by optimizing decoding performance. The diagnostic in Fig. S4 shows where this chosen upper-tail reference lies relative to quiet activity and to call-frame energy. It is a threshold diagnostic, not evidence that 99.9 is an empirically optimal percentile.

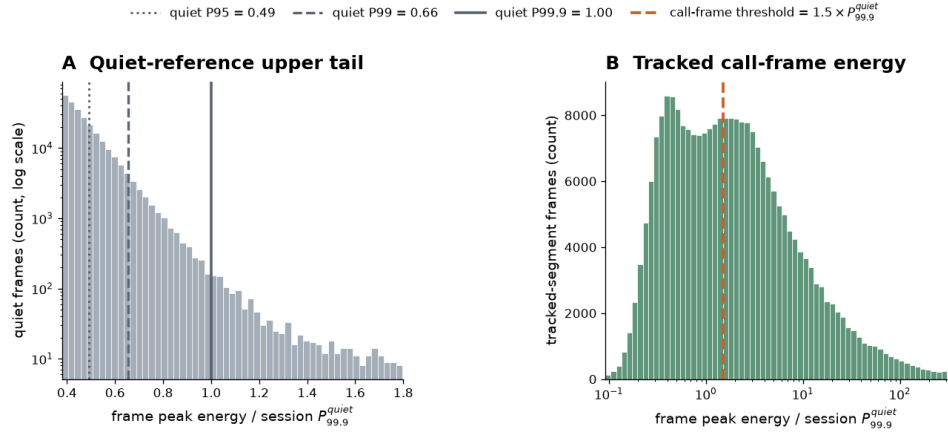


Figure S4: Quiet-reference threshold diagnostic. Panel A shows the upper tail of session-normalized quiet-period peak energy with P_{95} , P_{99} , and $P_{99.9}$ marked. Panel B shows tracked call-frame peak energies on a readable scale together with the final $1.5P_{99.9}^{quiet}$ threshold. The figure illustrates the chosen reference; it does not claim that 99.9 is an optimized percentile.

A.6 Call Clarity Criterion

For each trackable call, the clarity score q_i in Eq. (7) measures the fraction of tracked frames that clearly exceed the session’s quiet-period reference. The three reporting categories and the primary inclusion rule are defined once in Table 1 of the main article and are not repeated here. The primary clustering sample is exactly the union of the Clear and Partially Clear categories ($N = 2692$); Unclear/unusable detections are excluded before clustering.

The 1,292 Unclear/unusable detections consist of 849 low-SNR tracks, 256 calls with no reliable contour anchor, 182 edge-truncated contours, and 5 calls that are too short (Table S4).

Table S4: Reasons included in the Unclear/unusable category.

Reason	Calls
Low SNR ($q_i < 0.30$)	849
No reliable contour anchor	256
Edge-truncated contour	182
Too short	5
Total	1292

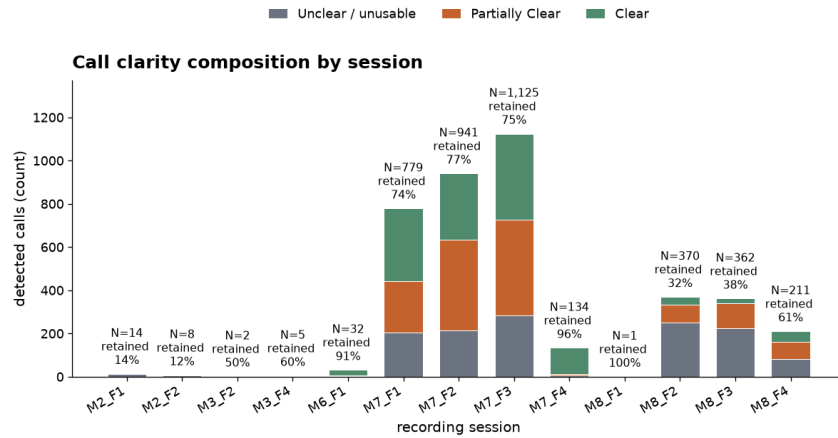


Figure S5: Session-wise acoustic clarity composition. Retention varies substantially across recordings, motivating the use of usable-call counts rather than raw detection counts for neural-decoding inclusion.

A.7 Does Acoustic Filtering Change the Sample?

The primary clustering intentionally uses only calls with reliable enough contours, but this filter can shift the acoustic distribution. Among excluded calls with measurable contours ($N = 1036$), the largest difference from retained calls is bandwidth (Cliff’s $\delta = -0.462$, medium effect). FM intercept and mean frequency show smaller shifts; most other features have negligible effect sizes. This does not invalidate clustering on the retained calls, but it limits how far the resulting categories should be generalized to removed detections.

Table S5: Retained versus excluded calls with measurable contours. Cliff's δ gives the effect size; BH-FDR is applied across the compared acoustic variables.

Feature	Retained median	Excluded median	Cliff's δ	p	FDR q	Effect
Duration (ms)	32.50	33.50	-0.059	0.0055	0.0062	negligible
FM intercept (kHz)	58.08	61.81	-0.238	1.61×10^{-29}	4.30×10^{-29}	small
FM slope (kHz/s)	59.73	34.67	0.050	0.0180	0.0180	negligible
Min frequency (kHz)	53.50	53.50	-0.065	0.0022	0.0030	negligible
Max frequency (kHz)	67.50	69.50	-0.142	1.67×10^{-11}	3.34×10^{-11}	negligible
Bandwidth (kHz)	15.45	18.04	-0.462	5.90×10^{-106}	4.72×10^{-105}	medium
Peak frequency (kHz)	62.00	63.50	-0.125	3.10×10^{-9}	4.97×10^{-9}	negligible
Mean frequency (kHz)	66.10	68.43	-0.330	6.44×10^{-55}	2.57×10^{-54}	small

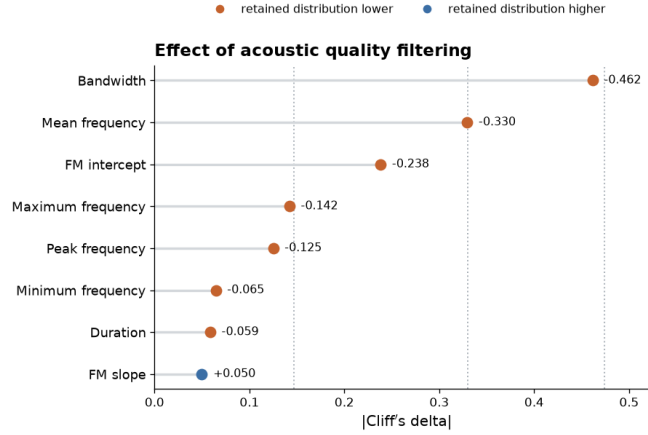


Figure S6: Effect of acoustic quality filtering on the retained sample. Points show Cliff's δ for each measurable acoustic variable; bandwidth has the largest shift ($\delta = -0.462$). This is a quality-control diagnostic rather than an independent biological result.

A.8 Bout Segmentation

The pooled inter-call interval distribution has one density peak near 0.14 s and no clear valley. We therefore choose the silence boundary from the knee of the number-of-bouts curve, obtaining 0.4 s. At this threshold, 79% of pooled inter-call intervals are assigned within a bout. The within-bout label-consistency score is 59.2%, indicating substantial local label structure. We therefore keep bouts intact during cross-validation; we do not interpret this score as an above-chance test because no dedicated null was used for this quantity.

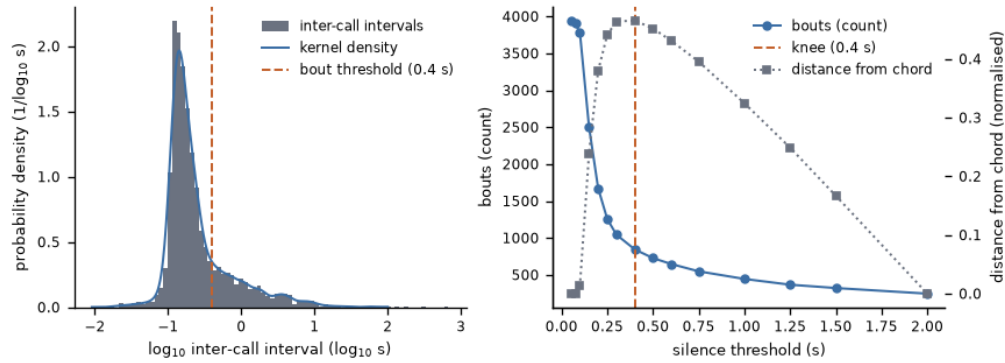


Figure S7: Bout-threshold selection. The 0.4-s threshold is derived from the knee of the bout-count curve because the inter-call interval distribution has no natural separating valley.

B. Acoustic Clustering Validation

B.1 Feature Correlation and Selection

Strong correlations are present among the candidate frequency variables (Table S6). The three-feature representation removes minimum and maximum frequency so that one frequency-related property is not counted multiple times by Euclidean distance.

Table S6: Strongly correlated candidate acoustic features.

Feature pair	Pearson r
FM intercept – minimum frequency	0.87
FM intercept – maximum frequency	0.88
Minimum – maximum frequency	0.86

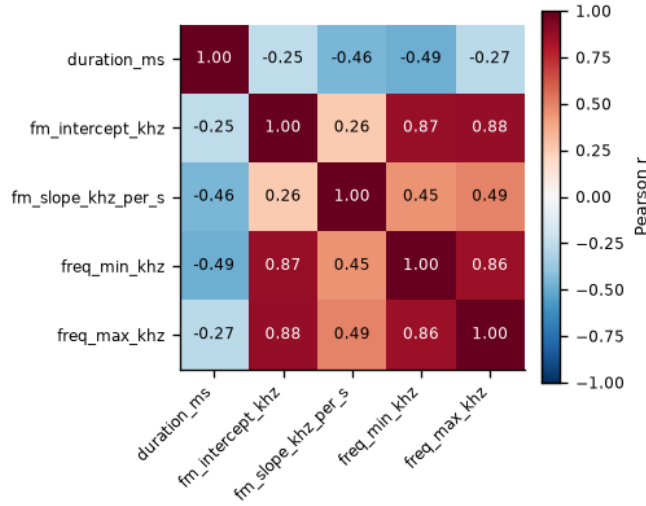


Figure S8: Correlation structure among candidate acoustic features. The three strongly correlated frequency variables motivate the reduced feature set.

B.2 Sensitivity to the Acoustic Feature Set

The reduced three-feature representation is not treated as interchangeable with the original five-feature set. Agreement between their K-Means partitions varies across K , from ARI = 0.385 to 0.910 (Table S7). The five-feature representation would select $K = 2$ by Silhouette, whereas the reduced representation selects $K = 3$. These Silhouette values are not compared directly across feature spaces because adding strongly correlated dimensions changes the Euclidean geometry itself.

Table S7: Agreement between K-Means partitions obtained from the five-feature and reduced three-feature acoustic representations.

K	ARI (five-feature vs. three-feature)
2	0.385
3	0.910
4	0.568
5	0.750
6	0.787
7	0.834

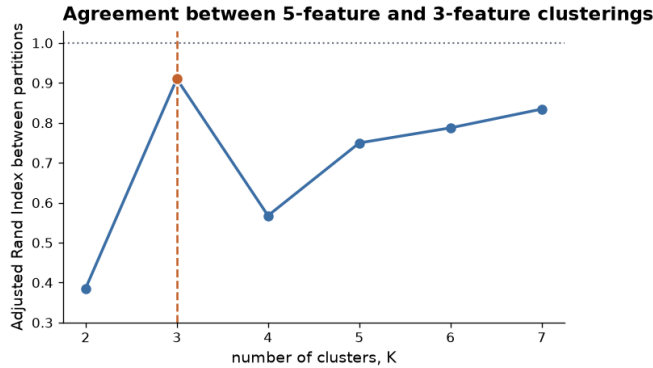


Figure S9: Sensitivity of the acoustic partition to feature-set choice. The high agreement at $K = 3$ does not hold uniformly across the full sweep, which is why the feature set is chosen from collinearity before K selection rather than from agreement at one selected K .

B.3 Full K -Selection Sweep

Table S8 reports all internal clustering metrics. Silhouette and Calinski–Harabasz select $K = 3$; Davies–Bouldin continues to improve to $K = 7$. Because Silhouette was defined as the primary criterion before the downstream neural analysis, $K = 3$ is retained. These indices evaluate the K-Means partitions across candidate values of K ; they are not used as a neutral head-to-head score between K-Means and GMM. Silhouette in particular rewards compact, separated distance-based clusters and is therefore naturally aligned with K-Means geometry. The GMM comparison is handled separately using BIC and partition agreement.

Table S8: Clustering metrics across candidate numbers of clusters. Arrows indicate the preferred direction of each internal metric.

K	Silhouette $\uparrow$	Davies–Bouldin $\downarrow$	Calinski–Harabasz $\uparrow$	Full-GMM BIC	Diag.-GMM BIC	KM/GMM ARI
2	0.377	1.077	1786.6	19815	20155	0.828
3	0.394	1.010	1791.8	17449	18246	0.693
4	0.365	0.960	1734.2	17002	17766	0.436
5	0.351	0.950	1717.6	16720	17452	0.489
6	0.358	0.951	1698.9	16589	17226	0.561
7	0.356	0.928	1665.3	16574	17151	0.539

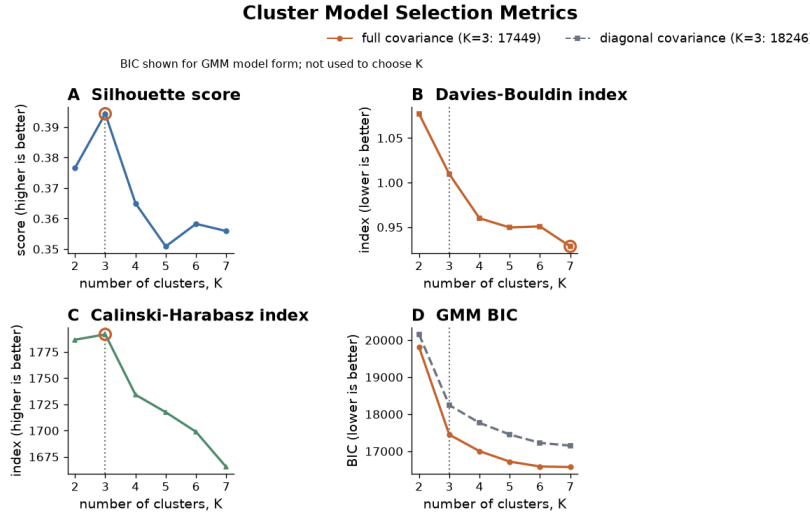


Figure S10: Selection of K and GMM fit diagnostics. Panels A–C show Silhouette (higher is better), Davies–Bouldin (lower is better), and Calinski–Harabasz (higher is better) for K-Means across $K = 2, \dots, 7$. Silhouette and Calinski–Harabasz select $K = 3$, while Davies–Bouldin favors $K = 7$. Panel D shows BIC for full- and diagonal-covariance GMMs; BIC is used to compare GMM model form and is not used to choose K because it decreases throughout the tested range.

B.4 K-Means and GMM Complexity

The three models use the same $d = 3$ acoustic features, but they estimate different numbers of quantities for each of the $K = 3$ clusters. K-Means estimates only the location of each cluster centroid, so

$$N_{\text{KMeans}} = Kd.$$

A diagonal-covariance GMM estimates a d -dimensional mean and d feature variances for every component, together with $K - 1$ free mixture weights:

$$N_{\text{GMM,diag}} = Kd + Kd + (K - 1).$$

A full-covariance GMM estimates a complete symmetric $d \times d$ covariance matrix for every component. Such a matrix contains $d(d + 1)/2$ independent entries, giving

$$N_{\text{GMM,full}} = Kd + K \frac{d(d + 1)}{2} + (K - 1).$$

For $K = 3$ and $d = 3$, these become

$$\begin{aligned} N_{\text{KMeans}} &= 3 \times 3 = 9, & N_{\text{GMM,diag}} &= 9 + 9 + 2 = 20, \\ N_{\text{GMM,full}} &= 9 + 18 + 2 = 29. \end{aligned}$$

The 18 covariance parameters in the full GMM come from six independent covariance terms per cluster: three feature variances and three pairwise covariances. The GMM has only $K - 1 = 2$ free mixture-weight parameters because the three component weights must sum to one. Within the GMM family, the full-covariance model is better supported at $K = 3$ by BIC (17,449 vs. 18,246 for diagonal covariance), meaning that the improvement in likelihood is sufficient to offset its additional parameters. However, the full-GMM and K-Means partitions still show substantial but not identical agreement ($\text{ARI} = 0.693$). Since the downstream analysis requires one hard acoustic label per call, we retain K-Means as the more parsimonious representation with a substantially lower parameter-estimation burden. Table S9 summarizes the trade-off.

Table S9: Model-complexity comparison at $K = 3$, $d = 3$. For K-Means, the count refers to estimated centroid coordinates; for GMMs, it refers to free distributional parameters.

Model	Parameters	Result
K-Means	9	selected
Diagonal GMM	20	BIC 18,246
Full GMM	29	BIC 17,449

B.5 Stability and Null Tests

The selected $K = 3$ solution has mean $\text{ARI} = 0.98$ across repeated 80% subsamples. The observed Silhouette (0.394) is also larger than all 200 shuffled-feature null draws (maximum 0.303) and all 200 single-Gaussian null draws with the same covariance (maximum 0.254), giving $p = 0.005$ for both tests. Cluster balance is 0.99.

Table S10: Primary clustering validation.

Check	Result
Selected K	3
Silhouette	0.394
80% subsampling ARI	0.98
Shuffled-feature null	$p = 0.005$
Single-Gaussian null	$p = 0.005$
Cluster balance score	0.99

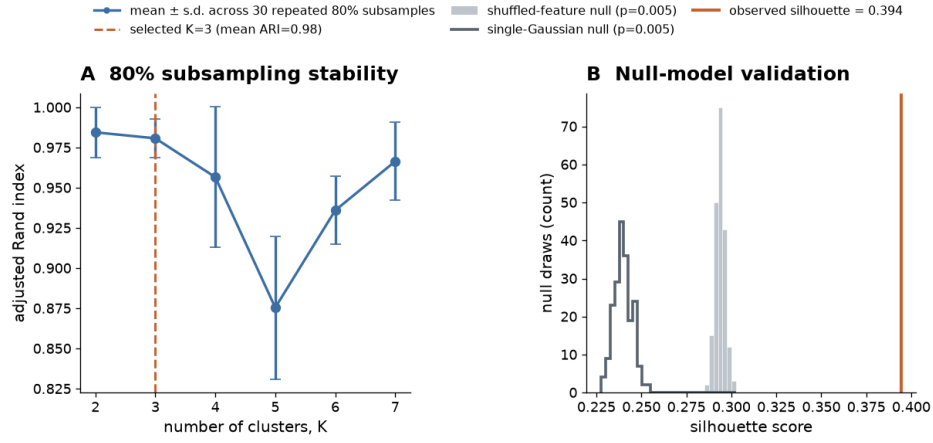


Figure S11: Stability and null-model validation. Panel A shows mean ARI $\pm$ s.d. across 30 repeated 80% subsamples for each K ; the selected $K = 3$ solution has mean ARI = 0.98. Panel B compares the observed Silhouette (0.394) with shuffled-feature and single-Gaussian null distributions; both empirical tests give $p = 0.005$.

B.6 Acoustic Cluster Profiles

The three operational categories differ strongly on the selected features. Pairwise Mann–Whitney comparisons with Holm correction and Cliff’s δ quantify the separations (Table S12). The largest effect is the FM-intercept difference between Clusters 0 and 2 ($\delta \approx 1.00$).

Table S11: Median acoustic characteristics of the three clusters.

Cl.	N	Dur. ms	Intercept kHz	Slope kHz/s
0	757	45.5	62.52	+16.67
1	994	18.0	60.70	+459.59
2	941	43.0	43.26	-75.94

Table S12: Pairwise acoustic comparisons. Cliff’s δ is an effect-size measure; Holm correction is applied within the pairwise testing family.

Feature	Clusters	Cliff’s δ	Holm p	Effect
Duration	0 vs. 1	0.856	1.61×10^{-206}	large
Duration	0 vs. 2	0.084	0.003	negligible
Duration	1 vs. 2	-0.773	8.60×10^{-190}	large
FM intercept	0 vs. 1	0.201	1.16×10^{-12}	small
FM intercept	0 vs. 2	1.000	1.52×10^{-274}	large
FM intercept	1 vs. 2	0.984	2.19×10^{-306}	large
FM slope	0 vs. 1	-0.902	2.32×10^{-229}	large
FM slope	0 vs. 2	0.365	7.46×10^{-38}	medium
FM slope	1 vs. 2	0.968	2.56×10^{-296}	large

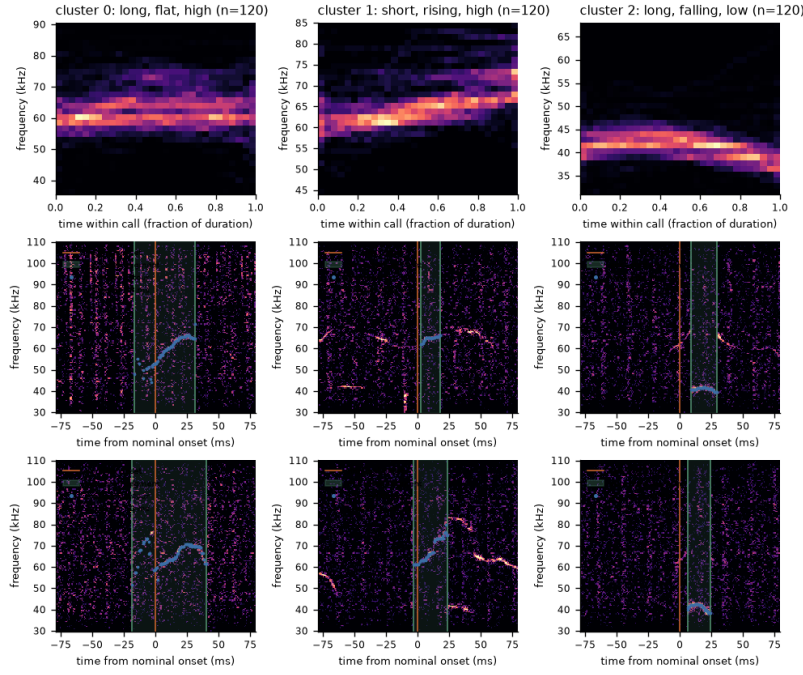


Figure S12: Representative acoustic structure of the three clusters. The top row summarizes pooled contour shape after duration normalization; lower rows show randomly selected calls at their true time scale.

B.7 Mouse Composition and Acoustic Distributions

The pooled clustering set is 84.4% M7, 14.3% M8, 1.1% M6, 0.15% M3, and 0.11% M2. The supplementary composition plot normalizes cluster fractions within each mouse and places them beside the distributions of duration, FM intercept, and FM slope. This helps separate two questions: whether mice contribute similar proportions of the three labels, and which measured acoustic properties define those labels. The smaller-mouse results remain descriptive because those animals contribute very few calls.

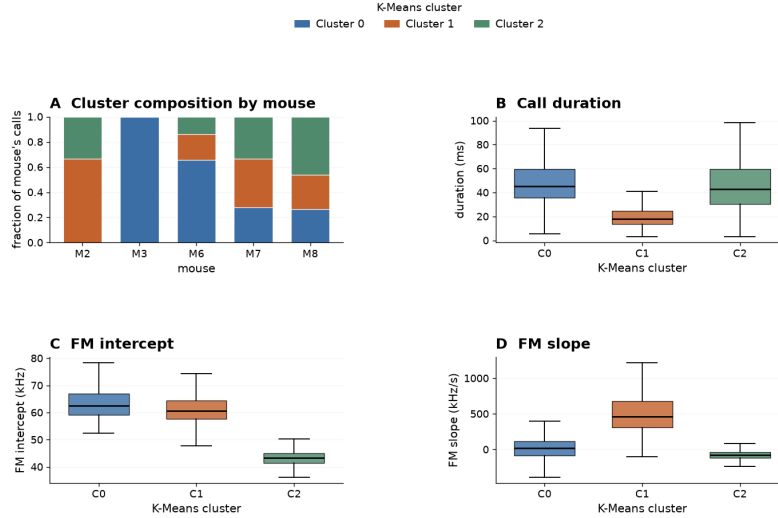


Figure S13: Cluster composition and acoustic feature distributions. The first panel shows cluster fractions normalized within each mouse; the remaining panels show the acoustic variables used to define the $K = 3$ partition.

B.8 Alternative Embeddings

PCA, t-SNE, and UMAP are used only to visualize the fixed K-Means labels; clustering is always performed in the original standardized three-feature space. Trustworthiness measures how well local neighborhoods from the original feature space are preserved in each embedding.

Table S13: Embedding trustworthiness ($k = 15$). Explained variance is defined for PCA only.

Embedding	Trustworthiness	Explained variance
PCA, 2-D	0.959	0.819
PCA, 3-D	1.000	1.000
t-SNE, 2-D	0.996	—
t-SNE, 3-D	0.998	—
UMAP, 2-D	0.991	—
UMAP, 3-D	0.995	—

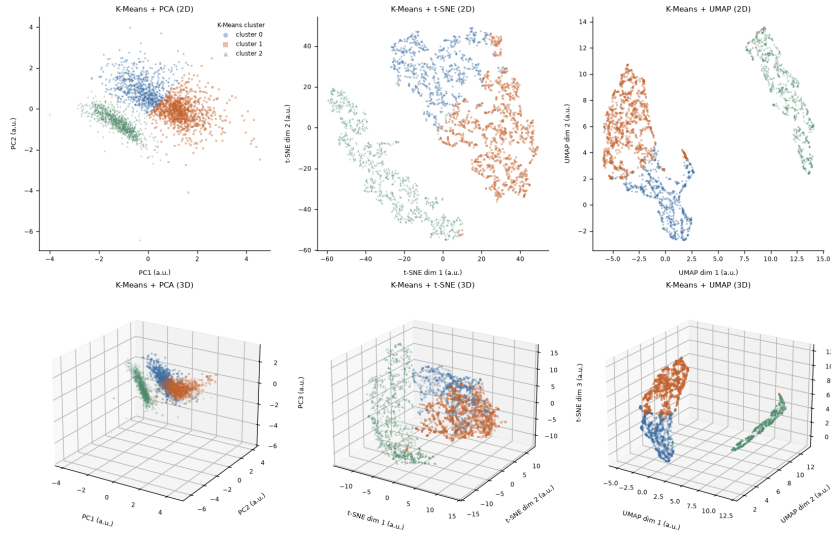


Figure S14: Low-dimensional visualizations of the acoustic categories using the fixed K-Means labels. These embeddings are descriptive only and do not determine cluster assignment.

For a direct visual comparison with the alternative clustering model, we also show the same family of embeddings colored by the hard labels from the final full-covariance GMM at $K = 3$. The GMM is fitted in the original standardized three-feature space; the PCA, t-SNE, and UMAP panels are descriptive visualizations and are not used to fit or select the model.

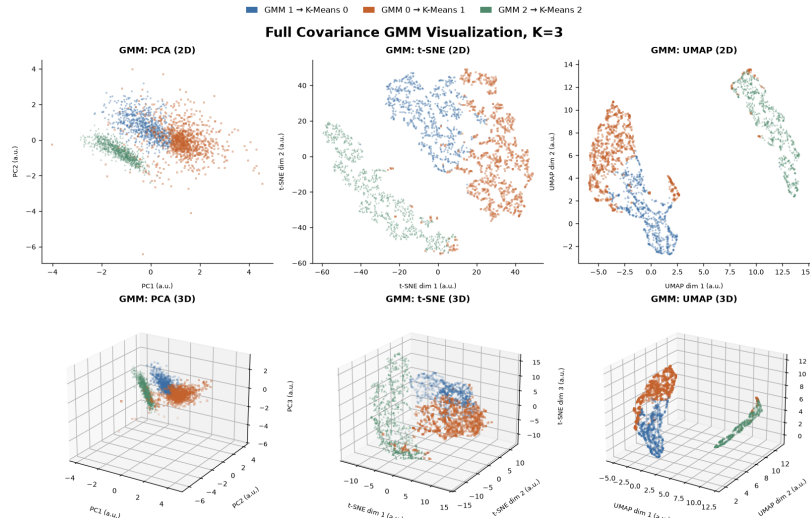


Figure S15: Alternative low-dimensional visualizations colored by the full-covariance GMM hard labels at $K = 3$. Panels show PCA, t-SNE, and UMAP in both 2-D and 3-D using the same standardized acoustic feature space. These embeddings are descriptive only; GMM fitting and hard-label assignment are performed in the original three-feature space.

B.8.1 Why do K-means and GMM look so similar?

The K-Means and GMM visualizations look very similar because both methods are applied to the same standardized acoustic features, and the embedding coordinates are identical between the two figures; only the cluster assignments change. After matching the arbitrary GMM component labels to their corresponding K-Means clusters (GMM 1 → **cluster 0 (blue)**, GMM 0 → **cluster 1 (orange)**, and GMM 2 → **cluster 2 (green)**), **88.2%** of calls retain the same assignment, while **319 of 2,692 calls (11.8%)** change cluster (Adjusted Rand index = 0.693). Thus, the two methods have a similar broad acoustic structure, but **not** the same partition.

As shown in Figure S16, most disagreements occur near the boundaries between clusters rather than in their cores. The largest exchange is between **cluster 0 (blue)** and **cluster 1 (orange)**, with 118 calls moving in one direction and 137 in the other. The **cluster 2 (green)** solution is more stable: 877 of its 941 K-Means calls remain together under GMM. Because these changes affect a minority of mostly boundary calls, the overall shapes of the two visualizations remain almost unchanged even though the partitions differ meaningfully.

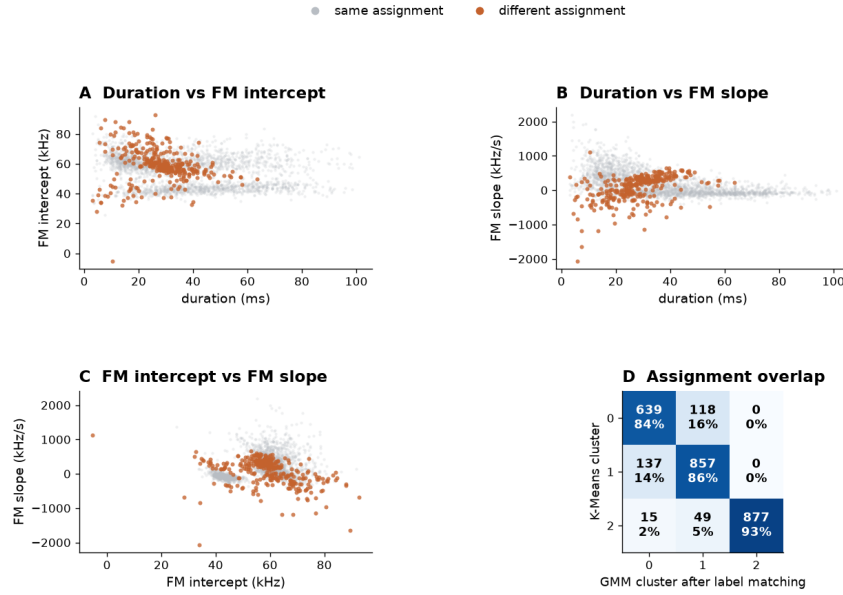


Figure S16: Comparison of K-Means and full-covariance GMM assignments. The two clustering methods recover the same broad organization of the three-feature acoustic space, but differ in the placement of some cluster boundaries. Calls assigned to corresponding clusters by both methods are shown separately from calls whose assignment changes after matching the arbitrary GMM component labels to the K-Means labels. The assignment matrix summarizes the overlap between the two solutions.

B.9 Acoustic Feature Ablation

Each primary acoustic feature is removed in turn and the remaining variables are re-clustered at $K = 3$. The resulting partition is compared with the full three-feature partition using ARI. This ablation asks how strongly each selected variable contributes to the final call-type definition: a lower ARI means removing that feature changes more cluster assignments. Removing FM intercept produces the largest change, followed by FM slope, whereas removing duration preserves most of the original assignments.

Table S14: Complete leave-one-feature-out acoustic clustering comparison. The full three-feature solution is included as the reference configuration.

Configuration	Features used	ARI vs. full
Full clustering	Duration + FM intercept + FM slope	1.000
Remove FM intercept	Duration + FM slope	0.334
Remove FM slope	Duration + FM intercept	0.446
Remove duration	FM intercept + FM slope	0.722

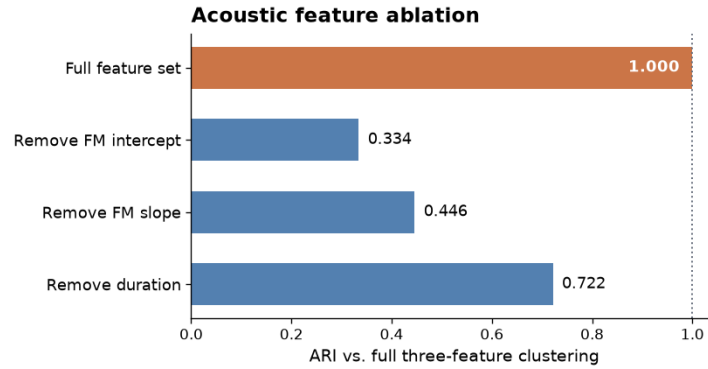


Figure S17: Acoustic feature ablation. Lower ARI indicates that removing the feature changes the $K = 3$ partition more strongly. FM intercept has the largest effect on the cluster assignments.

B.10 Clarity-Threshold Sensitivity

The acoustic categories are re-derived at three quality levels and compared on the primary sample. Clear-only clustering still agrees strongly with the primary labels (ARI = 0.861), while adding every otherwise valid tracked call gives ARI = 0.987.

Table S15: Re-clustering under different acoustic-quality thresholds.

Subset	Min. q_i	Calls	ARI
Clear only	0.60	1291	0.861
Primary	0.30	2692	1.000
All tracked	0.00	3541	0.987

C. Neural Decoding and Statistical Validation

C.1 Eligible Sessions and Class Balance

Seven sessions contain at least 100 usable labeled calls and enter neural decoding. Table S16 shows their class counts at the primary $K = 3$. Four of the sessions have a relatively small minority class, which motivates Macro-F1 and a separate exploratory minimum-class-size check.

Table S16: Primary $K = 3$ decoding samples.

Session	Calls	C0	C1	C2	Min.
M7_F1	574	146	202	226	146
M7_F2	727	224	321	182	182
M7_F3	841	249	340	252	249
M7_F4	129	13	18	98	13
M8_F2	119	22	27	70	22
M8_F3	136	63	58	15	15
M8_F4	129	17	19	93	17

C.2 Model Selection

The three classifier families are compared on identical grouped and temporally blocked outer folds. Hyperparameters are selected by three-fold inner cross-validation using training data only. **Macro-F1 is the model-selection score**; accuracy is not used as a second criterion for choosing the classifier family. Logistic regression has the best mean Macro-F1 and wins four of seven sessions, so it is carried forward. The most frequently selected logistic configuration across the 35 outer folds is $C = 0.01$ with 20 PCA components. After this comparison, that logistic-regression pipeline is fixed and used for both the observed-label and circular-null analyses.

Table S17: Complete nested-cross-validation model comparison. “Sessions won” counts sessions in which the model has the highest Macro-F1 among the three candidates.

Model	M7_F1	M7_F2	M7_F3	M7_F4	M8_F2	M8_F3	M8_F4	Mean F1	Sessions won
Logistic regression	0.370	0.369	0.370	0.345	0.416	0.356	0.286	0.359	4/7
Random forest	0.349	0.349	0.374	0.305	0.346	0.281	0.290	0.328	1/7
RBF-SVM	0.313	0.348	0.344	0.372	0.282	0.275	0.303	0.320	2/7

Table S18: Complete nested-search grid used during model comparison.

Component	Values tested
PCA components	5, 10, 20
Logistic C	0.01, 0.1, 1, 10
RF max. depth	None, 5, 10
RBF-SVM C	0.1, 1, 10

C.3 Full K -Sweep

The neural test is repeated for independently fitted acoustic partitions at $K = 2, 3, 4$. All 21 session-by- K tests belong to one BH-FDR family. Table S19 gives the full results.

Table S19: Neural decoding across $K = 2, 3, 4$. The only BH-FDR-significant configuration is M7_F2 at $K = 3$.

K	Session	Accuracy	Macro-F1	Circular p	BH q
2	M7_F1	0.589	0.509	0.5854	0.7995
2	M7_F2	0.508	0.506	0.2757	0.6105
2	M7_F3	0.520	0.520	0.4326	0.7570
2	M7_F4	0.791	0.531	0.3457	0.6599
2	M8_F2	0.664	0.564	0.1978	0.6105
2	M8_F3	0.478	0.475	0.6733	0.7995
2	M8_F4	0.713	0.464	0.7263	0.8027
3	M7_F1	0.378	0.333	0.6394	0.7995
3	M7_F2	0.399	0.392	0.0020	0.0420
3	M7_F3	0.373	0.370	0.1179	0.4951
3	M7_F4	0.705	0.406	0.0919	0.4825
3	M8_F2	0.479	0.359	0.2847	0.6105
3	M8_F3	0.346	0.291	0.6853	0.7995
3	M8_F4	0.527	0.272	0.8811	0.9252
4	M7_F1	0.275	0.258	0.5455	0.7995
4	M7_F2	0.292	0.280	0.0250	0.2622
4	M7_F3	0.308	0.284	0.0669	0.4685
4	M7_F4	0.318	0.217	0.6503	0.7995
4	M8_F2	0.286	0.264	0.2847	0.6105
4	M8_F3	0.301	0.256	0.2907	0.6105
4	M8_F4	0.209	0.160	0.9580	0.9580

C.4 Look-Back Window Sensitivity

The pre-call look-back is swept over 0.5, 1, and 2 s while keeping 0.5-s sub-bins. Only one of 21 window-by-session tests survives BH-FDR, again M7_F2 for the 1-s setting. The 1-s window also has the highest mean performance, but the sweep is treated as a robustness analysis rather than as post-hoc model selection.

Table S20: Macro-F1 by pre-call window. Bold marks the only corrected significant configuration in this 21-test family.

Look-back	M7_F1	M7_F2	M7_F3	M7_F4	M8_F2	M8_F3	M8_F4
0.5 s	0.336	0.369	0.353	0.331	0.345	0.312	0.325
1.0 s	0.333	0.392	0.370	0.406	0.359	0.291	0.272
2.0 s	0.313	0.399	0.386	0.331	0.383	0.223	0.361

C.5 Control Analyses

The calling-vs.-quiet control uses half-second windows labeled by whether they contain a call; the post-call type control uses the same $K = 3$ acoustic labels but moves the neural window immediately after call onset. No individual control survives BH-FDR correction (Table S21).

Table S21: Per-session control results. Reported p -values are circular-shift values before correction; neither control has an individually significant session after BH-FDR.

Session	Calling/quiet F1	p	Post-call type F1	p
M7_F1	0.603	0.0150	0.387	0.3447
M7_F2	0.557	0.0509	0.365	0.0739
M7_F3	0.552	0.0839	0.353	0.2897
M7_F4	0.566	0.0619	0.366	0.2727
M8_F2	0.499	0.4226	0.355	0.4016
M8_F3	0.495	0.5774	0.332	0.4056
M8_F4	0.513	0.0519	0.341	0.3636

C.6 Combined Evidence Across Sessions

Fisher’s and Stouffer’s methods provide a secondary summary of evidence pooled across recording sessions. BH-FDR is applied separately across the three call-type targets and across the two controls. Only calling-vs.-quiet survives correction (Table S22). Because several sessions come from the same mice, this pooled analysis should not be interpreted as independent animal-level replication.

Table S22: Secondary pooled tests across sessions.

Target	Fisher p	Stouffer p	BH q
Call type, $K = 2$	0.5967	0.3891	0.5967
Call type, $K = 3$	0.0267	0.0681	0.0800
Call type, $K = 4$	0.1324	0.1871	0.1986
Post-call type	0.2037	0.0732	0.2037
Calling vs. quiet	0.0024	0.0008	0.0047

C.7 Descriptive Pre-Call Population Activity (PSTH)

A peri-stimulus time histogram (PSTH) aligns repeated events to a common reference time and summarizes how mean neural activity changes around that event. Here the event is call onset. For this descriptive visualization only, each neuron’s `spikeCounts100ms` trace is z-scored within its recording session and the 1 s before each call is divided into 100-ms bins. Each call is then baseline-corrected to its own early pre-call interval (−1.0 to −0.7 s).

The PSTH is intentionally different from the decoder. The classifier uses cell-specific features from two 0.5-s bins and can exploit multivariate patterns across neurons. The PSTH first averages across neurons, so neuron-specific increases and decreases can cancel. Therefore a larger separation of the PSTH curves is not required for better decoding. In the single-session comparison, M7_F2 ($q = 0.042$) has more compact population-average traces than M7_F1 ($q \approx 0.800$). This is not evidence that M7_F2 is “worse” neurally; it instead shows that the significant decoding result cannot be explained simply by larger mean population activation.

For the single-session PSTHs, calls are first averaged within bouts and the shaded band is ± 1 SEM across bout-level traces. For the pooled PSTH, every recording contributes one cluster mean and the shaded band is ± 1 s.d. across session means. All PSTH panels use the same normalization and y-axis scale, so their apparent compactness can be compared descriptively.

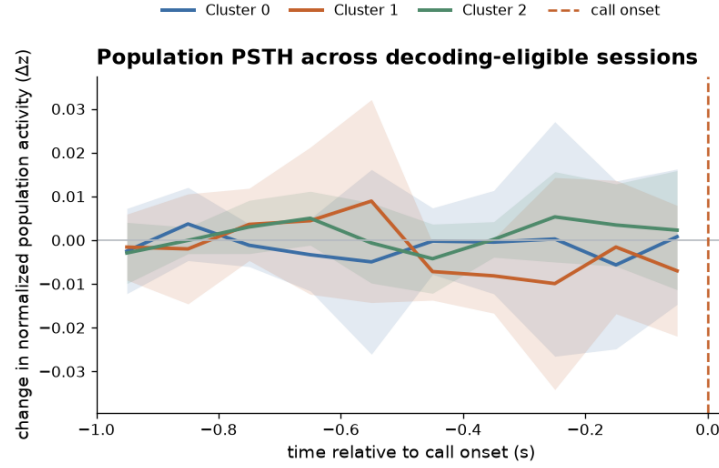


Figure S18: Population PSTH across decoding-eligible sessions. Calls are aligned to onset ($t = 0$), neural activity is summarized in 100-ms bins over the preceding second, and each call is baseline-corrected to -1.0 to -0.7 s. Solid lines are means across session-level cluster traces; shaded regions are ± 1 s.d. across session means. The figure is descriptive and is not the feature representation used by the classifier.

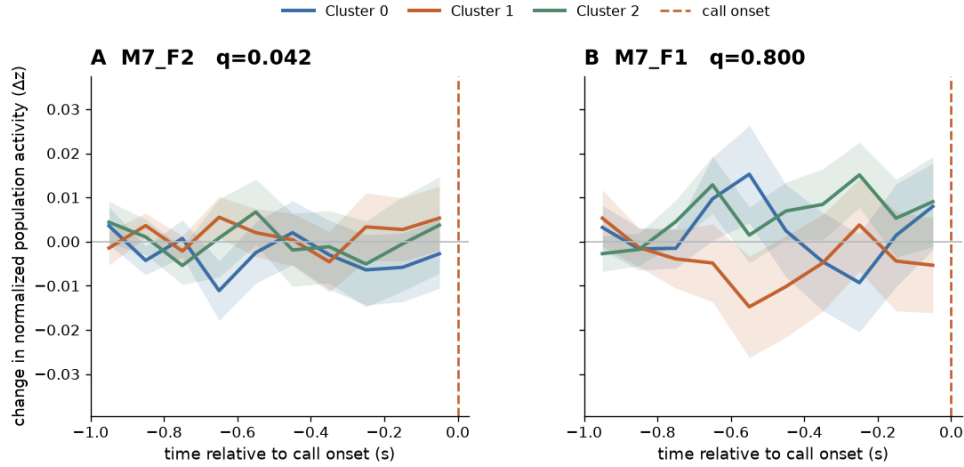


Figure S19: Session-level PSTH comparison. M7.F2 (left, $q = 0.042$) is the only significant $K = 3$ decoding session, while M7.F1 (right, $q \approx 0.800$) is non-significant. Lines are mean bout-level traces for each acoustic cluster and shading is ± 1 SEM across bouts. Both panels use identical normalization and y-axis limits. The more compact M7.F2 traces show that successful multivariate decoding does not require larger separation in the population-averaged PSTH.

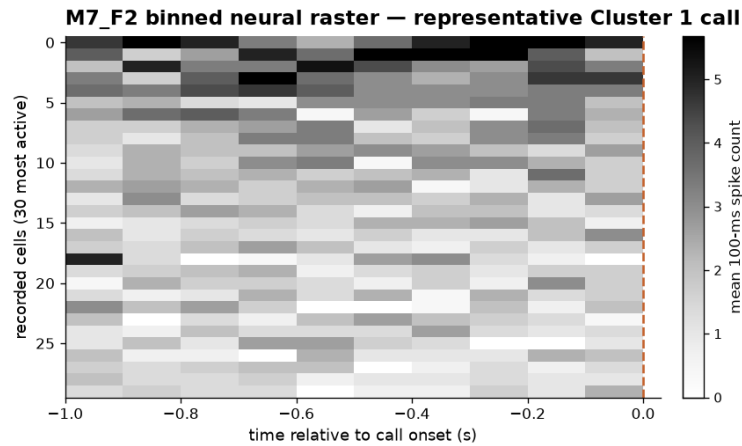


Figure S20: Representative M7_F2 binned neural raster. The 30 most active recorded cells are shown for one representative call using the available 100-ms spike-count representation. The dashed line marks call onset. The heterogeneous cell-level patterns illustrate information that can be lost when activity is collapsed into a single population mean.

D. Robustness and Ablation Tests

D.1 Recorded Brain Region

The recording composition differs strongly between M7 and M8. Cells are grouped as PFC when their area label begins with PL, ILA, or ACAd; all other labels are grouped as Other. M7 contains mostly PFC cells, whereas M8 contains mostly non-PFC cells (Table S23). This

imbalance means cell group is partly confounded with animal identity.

Table S23: Recorded-cell counts by broad region.

Session	All	PFC	Other
M7_F1	166	150	16
M7_F2	177	158	19
M7_F3	163	144	19
M7_F4	212	165	47
M8_F2	130	35	95
M8_F3	98	17	81
M8_F4	139	37	102

Mean Macro-F1 across sessions is 0.346 using all cells, 0.361 using PFC cells, and 0.314 using Other cells. Only one of 21 region-by-session tests is significant after correction within this ablation family, so the data do not support a strong region-specific conclusion.

Table S24: Macro-F1 by broad neural cell group.

Group	M7_F1	M7_F2	M7_F3	M7_F4	M8_F2	M8_F3	M8_F4
All	0.333	0.392	0.370	0.406	0.359	0.291	0.272
PFC	0.336	0.383	0.389	0.393	0.368	0.286	0.373
Other	0.388	0.334	0.351	0.250	0.342	0.281	0.249

D.2 Minimum Per-Class Sample Size

The predefined inclusion rule uses total usable calls rather than a minimum per class. As a post-hoc sensitivity check, we retain only sessions with at least 30 examples in every $K = 3$ cluster. Only M7_F1, M7_F2, and M7_F3 satisfy this criterion.

Table S25: Class counts used for the post-hoc minimum-class-size check.

Session	C0	C1	C2	Minimum
M7_F1	146	202	226	146
M7_F2	224	321	182	182
M7_F3	249	340	252	249
M7_F4	13	18	98	13
M8_F2	22	27	70	22
M8_F3	63	58	15	15
M8_F4	17	19	93	17

Within the restricted three-session family, M7_F2 remains significant at $K = 3$ ($q = 0.018$). Because the threshold was examined after seeing the main results and all retained sessions come from M7, this analysis is exploratory and does not replace the primary conclusion.

Table S26: Restricted-session primary $K = 3$ results.

Session	Macro-F1	Restricted q
M7_F1	0.333	0.6394
M7_F2	0.392	0.0180
M7_F3	0.370	0.2652

D.3 Bout-Threshold Sensitivity

The primary bout gap is 0.4 s. Repeating the $K = 3$ decoding with 0.3- and 0.6-s gaps leaves the overall conclusion unchanged. Mean Macro-F1 is 0.351, 0.355, and 0.344 respectively.

Table S27: Mean Macro-F1 by bout threshold.

Bout threshold	Mean Macro-F1
0.30 s	0.351
0.40 s	0.355
0.60 s	0.344

D.4 Effect of Bout Leakage: Grouped versus Random Cross-Validation

To show why bout grouping is necessary, we intentionally repeat the primary $K = 3$ analysis with invalid random K-fold splitting over individual calls. This split allows calls from the same bout, which can have overlapping neural windows and related labels, to appear in both training and test data. The random-CV values are therefore a leakage demonstration, not valid decoding results.

Table S28: Apparent performance inflation when bout grouping is ignored.

Session	Grouped F1	Random F1	Inflation
M7_F1	0.333	0.468	+0.134
M7_F2	0.392	0.407	+0.015
M7_F3	0.370	0.396	+0.026
M7_F4	0.406	0.418	+0.012
M8_F2	0.359	0.459	+0.100
M8_F3	0.291	0.323	+0.031
M8_F4	0.272	0.346	+0.074

Random CV increases Macro-F1 by 0.056 on average, with session-level inflation from +0.012 to +0.134. This directly supports the bout-grouped validation used in the main analysis.

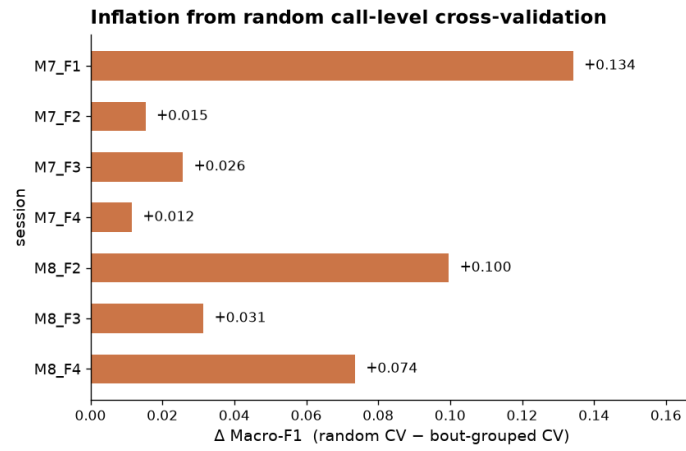


Figure S21: Performance inflation caused by ignoring bout structure. Random-CV results are shown only as a leakage demonstration.